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Noise-Robust and Edge Deployable ML Framework for Predictive Maintenance of Industrial Motors

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***Abstract*—Industrial three-phase induction motors are critical to manufacturing operations, yet their failure can cause costly downtime and maintenance overhead. This work introduces a machine learning framework tailored for fault diagnosis and severity assessment in such motors, using synthetic data that replicates six realistic electromagnetic and mechanical fault conditions. The system employs synchronized vibration and current signals, and evaluates several classifiers—including LSTM, GRU, TCN, RNN, and Random Forest—under varying industrial noise profiles such as phase jitter, frequency drift, and transient spikes. Experimental results demonstrate that time-series models consistently outperform classical approaches in noisy environments. For deployment on resource-constrained edge devices, structured pruning and quantization are applied to reduce model size and latency. Beyond motor fault classification, the framework also detects and visualizes approximately spherical clusters to comprehend fault severity levels. The proposed system enables scalable fault monitoring and low-latency predictive maintenance optimized for embedded industrial environments.**

***Index Terms*—Edge AI, edge computing, induction motor, industrial IoT, predictive maintenance, synthetic data**

I. Introduction

Three-phase induction motors are foundational to industrial operations, manufacturing processes and transportation infrastructure worldwide [1]. They are used to power a vast array of applications due to their robustness, ease of maintenance, and high efficiency. Given their crucial role, their uninterrupted operation is critical to production continuity. Even minor performance degradations can result in significant financial losses, production downtime, and potential safety hazards. Consequently, there is growing interest in predictive maintenance strategies that can detect faults early, reduce unplanned outages, and improve overall operational efficiency [2].

Traditionally, diagnostic methods for industrial motors have relied on scheduled maintenance checks, simple threshold-based monitoring approaches and human expertise for analysis [3]. While these approaches provide some fault coverage, they are often reactive, lack sensitivity to early-stage anomalies, depend heavily on domain expertise and are prone to human errors. With advancements in sensor technologies, data acquisition and computational capabilities, machine learning (ML) techniques have emerged as a powerful alternative for automating fault detection. ML algorithms can extract subtle fault signatures, adapt to varying operating conditions, and offer more accurate and timely diagnostics [4]. By analyzing patterns in operational data, these algorithms can detect anomalies and infer the health state of an induction motor more systematically and with greater precision and with minimized human intervention [3].

Despite their promise, training robust ML models for industrial environments presents several challenges. First, the development of robust models requires diverse, high-quality datasets representing various fault scenarios. However, collecting labeled data under all possible fault conditions is expensive, time-consuming, and often unsafe, especially if it involves deliberately inducing faults in operational machinery [5]. Second, industrial signals are subject to high noise levels, phase jitters, and intermittent interruptions resulting from voltage instabilities or environmental factors. These noise sources can hamper the performance of conventional algorithms, which struggle to extract reliable features under real-world conditions [6]. Third, real-time diagnostics demand continuous processing of high-throughput sensor streams, often on resource-constrained edge devices with limited memory, compute power, and energy availability. These constraints necessitate lightweight and compressed ML models suitable for embedded deployments for onsite monitoring [7].

In this study, we present a comprehensive framework for fault diagnosis in industrial three-phase induction motors, covering the entire diagnostic pipeline from data collection to deployment in resource-limited embedded systems. Our approach focuses on six distinct electromagnetic and mechanical faults and evaluating multiple architectures under varying industrial noise conditions. To enhance real-world applicability, we apply compression techniques to increase efficiency and support embedded deployment. Furthermore, we extend our analysis beyond fault classification by employing cluster visualizations to assess fault severity levels, providing a holistic understanding of fault progression and motor health.

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II. Literature Review

Machine learning-based motor fault diagnosis has evolved significantly, offering advancements in predictive maintenance strategies for industrial three-phase induction motors. By automating feature extraction and improving classification accuracy, ML techniques have outperformed traditional rule-based and threshold-driven approaches. Classical models such as Support Vector Machines (SVM) [8] and modern deep learning models like Convolutional Neural Networks (CNNs) [9] have been successfully applied to machine condition monitoring and fault diagnosis, with CNNs demonstrating efficacy in processing vibrational spectroscopic data and eliminating the need for manual preprocessing. Similarly, deep transfer learning has been employed to improve diagnostic precision. A transfer learning-based framework is proposed in [10] that converts sensor data into time-frequency representations, allowing for superior classification performance across different mechanical components.

One of the primary challenges in machine learning for fault diagnosis is the availability of sufficient labeled data. Generative Adversarial Networks (GANs) have been explored as a viable solution for augmenting limited datasets. In [11], it is demonstrated how GANs could synthesize high-fidelity training data, thereby improving model generalization. Additionally, hybrid approaches that integrate physics-based modeling with deep learning have been proposed to enhance decision-making processes. A physics-informed neural networks is proposed in [12] that leverage domain knowledge to refine fault detection models. Since motor fault signals are inherently sequential, time-series analysis plays a crucial role in fault diagnosis. Recurrent Neural Networks (RNNs) have been widely adopted to capture temporal dependencies in signal measurements. The authors in [13] reviewed various RNN architectures, including Long Short-Term Memory (LSTM) networks, emphasizing their ability to model transient fault patterns in motor vibration and current signatures. These architectures have proven effective in diagnosing faults even in noisy environments.

Recent advancements in Embedded AI have enabled real-time fault diagnosis on resource-constrained edge computing devices. A comprehensive review of AI integration is provided in [7] for embedded systems, demonstrating the feasibility of deploying sophisticated machine learning models in low-power environments. This innovation is particularly beneficial for industrial settings where continuous monitoring is required with minimal computational overhead. Attention-based architecture has been increasingly explored to enhance the detection of localized signal irregularities. In [14], an attention-driven fault diagnosis system is explained that selectively weighed relevant signal segments, thereby improving classification accuracy. Zhou et al. [15] further expanded on this concept by integrating attention mechanisms with hybrid CNN-MLP models, effectively detecting sporadic and noise-obscured fault signatures in motor systems.

A. Research Gap

Although the aforementioned studies mention significant contributions, it is important to highlight that much of the existing studies focus on specific model types under controlled conditions. Usually, a narrowly selected aspect of motor behavior is examined, such as stator failures, or rotor breakage analysis. Also, studies more often utilize a binary classification approach to assess if the motor is healthy or faulty, instead of incorporating more extensive multiclass situations. Another important need is to comprehensively model the noise conditions, since results derived from controlled laboratory settings may not always correspond with actual industrial environments. Finally, it can be stated that models that perform well frequently remain untested in embedded systems or constrained deployment environments, which are the most crucial in industrial setups.

B. Our Contribution

Our study addresses identified gaps by examining six multiclass cases covering electromagnetic and mechanical faults. The evaluation of the developed models is conducted within a comprehensive noise modeling framework that involves gradually increasing frequency drift, phase jitter, background Gaussian noise, and amplitude distortions. This approach discusses the progression of model performance as the electromagnetic environment becomes increasingly complex. We then apply quantization and pruning to optimize Edge AI feasibility. Also, our work goes beyond mere classification, and provides approximately spherical clustering results of spectral and frequency features, representing the severity and progression of faults.

III. Methodology

A. Synthetic Data Generation

This work builds upon our prior study [16], where we developed a high-fidelity synthetic signal generation framework for replicating the behavior of industrial three-phase induction motors under multiple fault conditions. The dataset based on real motor data [17] models a four-pole, three-phase induction motor (AGM2E 112 M 4a) operating at the European standard of 50 Hz, with a rated voltage of 400 V and a nominal speed of 1455 RPM. Such motors are commonly deployed in conveyor systems, HVAC compressors, and water treatment pumps due to their robustness and efficiency.

In this study, six operational states are considered: Healthy, Rotor Bar Defect, Stator Winding Fault, Current Overload, Insufficient Current, and Mechanical Fault. Each fault type represents a distinct electrical or mechanical disruption, leading to characteristic patterns in current and vibration signals. These effects are observable in both time and frequency domains [18] These fault-specific signal characteristics are outlined in Table I.

To replicate these conditions synthetically, we used a hybrid approach combining statistical modeling, signal transformation, and domain-specific waveform formulations, as detailed in [16]. The vibration data were generated using

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TABLE I
INDUCTION MOTOR STATES AND CORRESPONDING VIBRATION AND CURRENT BEHAVIOR

State Name	Vibration Behavior	Current Behavior
Healthy State	Stable amplitudes	Optimal 120 degrees between three phase angles with even amplitudes
Rotor Bar Defect	Unstable and elevated	Sideband frequencies are observed due to broken rotor bars, distorted phases, and amplitude modulation
Stator Winding Fault	Unstable and elevated	Even harmonics are observed due to uneven magnetic field, distorted phases
Current Overload	Higher amplitudes	Elevated phase amplitudes
Insufficient Current	Lower amplitudes	Decreased phase amplitudes
Mechanical Fault	Unstable and elevated	Slight amplitude and phase fluctuations due to torque ripple and uneven load transfer

probabilistic models fitted to real-world datasets and validated via clustering-based behavioral analysis. For current signals, sinusoidal waveforms were synthesized using analytical expressions for each phase, with specific modulations (e.g., harmonic injection for stator faults or sideband effects for rotor bar defects) applied to emulate realistic fault conditions.

The training dataset was generated at a 100 kHz sampling rate, with overlapping 1000 sample windows (10 ms) designed to capture transient fault signatures rather than full-cycle harmonics. Configurable noise sources, including phase jitter, frequency drift, Gaussian interference, and micro-interruptions were added to increase realism. Each sample contained synchronized three-phase current and three-axis vibration data across randomized motor state sequences. To prevent overly deterministic transitions, the order and sequence lengths of motor states were randomized at each run. This resulted in a multi-gigabyte training dataset that comprises 240,000 overlapping windows with 50% healthy (120,000) samples and 50% faulty (120,000) samples distributed among the five fault classes in the proportions shown in Fig. 1. The validation set follows the same class distribution as the training data. To enable class-fair evaluation and clear confusion-matrix interpretation, separate balanced test datasets of 60,000 samples (10,000 per class) were generated under both low- and high-noise conditions, ensuring no overlap with the training data. Each test sample similarly contained tri-axial vibration and three-phase current signals at a 100 kHz sampling rate, enabling unbiased benchmarking of the classification models.

B. Feature Engineering

To enable efficient and real-time classification, we apply a feature engineering pipeline that transforms the raw signals into compact, informative descriptors. Each synchronized current and vibration stream is sampled at 100 kHz and divided into overlapping sliding windows of 1000 samples, targeting transient signatures rather than complete waveform cycles. For every window, a total of 78 time-domain and frequency-domain features are computed across the three accelerometer

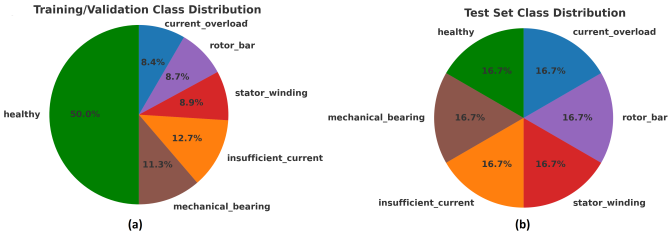


Fig. 1. (a) Training/validation class distribution (b) Test dataset class distribution

axes and three-phase current signals. Key time-domain metrics include root mean square (RMS), standard deviation, skewness, and kurtosis for capturing energy bursts, signal dispersion, and impulsiveness. Frequency-domain attributes include dominant frequency, sideband amplitude strength (linked to rotor faults), and even-harmonic energy (indicative of stator anomalies). Spectral entropy is also used as a measure of complexity to characterize signal irregularities.

The resulting feature matrix forms a structured, low-dimensional representation of the high-rate sensor data. This not only reduces memory and computation overhead but also makes the data suitable for deployment in traditional ML or temporal DL architectures such as LSTM, GRU, or TCN. As illustrated in Fig.2, this representation supports real-time inference by enabling near real-time feature computation and immediate model input.

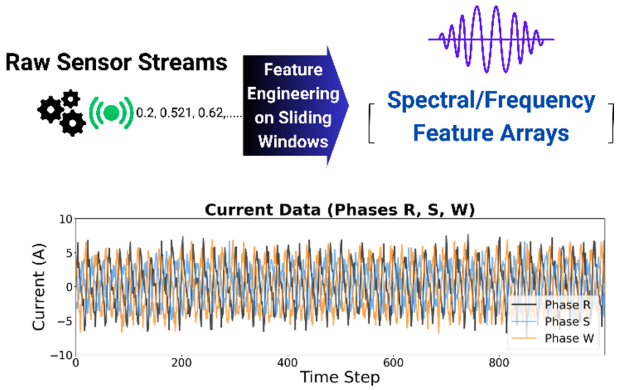


Fig. 2. System flow to obtain feature arrays before feeding them into the AI/ML stage.

C. Model Evaluation under Varying Noise Levels

To evaluate the robustness of machine learning models in industrial motor fault diagnosis, we examined both classical and time-series-based approaches across both low and high noise levels. The primary objective was to assess the capability of each architecture to handle noisy and sequential signal features typical of real-world industrial environments.

We considered five model types in total. Random Forest (RF) was selected as the baseline classical approach, operating on individual feature vectors without incorporating temporal context. In contrast, the remaining four models—Recurrent

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Neural Network (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Network (TCN)—were chosen for their ability to model temporal dependencies within sliding windows of time-series data. LSTM and GRU models are specialized RNN variants that address vanishing gradient issues and are well-suited for capturing long-term dependencies in motor signal sequences [19]. TCN offers a parallelizable alternative by leveraging causal convolutions and dilation [20].

All deep learning models use a lookback window of $n = 15$, i.e., each prediction uses $[x_{t-14}, \ldots, x_t]$ (current + previous 14 feature vectors). A summary of the models and their configurations is presented in Table II.

TABLE II
MODEL CONFIGURATION/HYPERPARAMETER SUMMARY

Model	Configuration Details
LSTM / GRU / RNN	64 hidden units, dense-head ReLU, lookback window $n = 15$, Adam optimizer, early stopping, categorical cross-entropy loss
TCN (Low Noise)	Kernel size = 3, Dilation = [1,2,4,8], causal padding, dropout = 0.2
TCN (High Noise)	Kernel size = 2, Dilation = [1,2,4,8,16,32], causal padding, dropout = 0.2
Random Forest	200 estimators, Gini impurity criterion, majority voting aggregation

The TCN configuration differs between the low-noise and high-noise settings. For low-noise data, a *larger kernel with smaller dilation* was sufficient to capture localized temporal patterns. In contrast, high-noise conditions benefited from a *smaller kernel with larger dilation*, which expands the receptive field to aggregate longer-term context and improves robustness to transient anomalies.

To mimic industrial signal degradation, we adopt a progressive noise-injection framework. For *mild* interference, disturbance parameters are kept small (low AM depth α, low noise variance σ^2, small baseline-drift slope d, infrequent/short dropouts via $m(t)$, and small time-warp magnitudes in $\tau(t)$). For *high* interference, these parameters are increased to reflect severe and variable operating conditions. The noisy signal $y(t)$ is generated as

$$y(t) = m(t)\left(1+\alpha\sin(2\pi f_m t)\right)x\big(t+\tau(t)\big) + b_0 + d\,t + w(t), \tag{1}$$

where $m(t) \in \{0, 1\}$ is a dropout mask and

$$\tau(t) = \tau_{\text{jitter}}(t) + \int_0^t \gamma(\lambda)\,d\lambda$$

encodes phase jitter and frequency drift. Here α is the amplitude-modulation depth (dimensionless), f_m is the AM frequency [Hz], $w(t) \sim \mathcal{N}(0, \sigma^2)$ is additive white noise (same units as x), b_0 is a DC offset, and d is the baseline-drift slope [units/s].

Additionally, we accounted for transient operating states where motors briefly shift between normal and faulty behavior. These transitions often manifest as short-lived deviations that can be misclassified by traditional algorithms. The time-series models were designed to mitigate this challenge by aggregating historical signal context across the lookback window, enabling the detection of persistent fault signatures even in the presence of abrupt or partial disruptions. This evaluation framework supports the identification of models that are not only accurate under clean laboratory signals but also resilient to the noise patterns commonly found in deployed industrial systems. Although our synthetic noise modeling captures jitter, drift, Gaussian interference, and micro-interruptions, real industrial environments also exhibit harmonic distortions, load fluctuations, and thermal effects [21]. Incorporating these in future work will improve realism.

To evaluate the classification performance of our models across all fault categories, we employed the *Macro F1-score* as the primary metric. Unlike micro-averaged metrics that are influenced by class imbalance, the macro F1-score treats all classes equally by computing the F1-score independently for each class and then averaging them. This makes it particularly suitable for multi-class fault diagnosis tasks where each class's performance is equally important. The F1-score for a single class is defined as the harmonic mean of its precision and recall as shown in (2)

$$\text{F1}_i = \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{2}$$

The macro F1-score is then calculated as in Eq. (3)

$$F1_{\text{Macro}} = \frac{1}{C}\sum_{i=1}^{C} F_{1,i} \tag{3}$$

where C is the total number of classes and F1_i is the F1-score for class i. This metric ensures that each class contributes equally to the overall evaluation, providing a fair measure of model performance across all fault types.

D. Optimization for Low-Power Constrained Environments

To address the resource constraints often encountered in industrial environments [22], advanced compression techniques for Edge AI has gained momentum [23] to allow inference directly on local, lower-power boards. To support real-time fault diagnosis in industrial environments, we applied embedded machine learning optimization techniques to reduce model size and inference latency. These optimizations enable the deployment of trained models on edge AI platforms with limited computational and memory resources. An example server-to-edge transformation diagram can be seen in Fig. 3

The methodology focuses on two key techniques: *structured pruning* and *post-training quantization*. Structured pruning was implemented using the TensorFlow Model Optimization Toolkit (TFMOT), where low-importance channels/units were progressively removed during training via a controlled sparsity schedule. After pruning, the corresponding channels/units were physically removed and the model was fine-tuned to recover any lost accuracy. The pruning operation can be formally described as (4):

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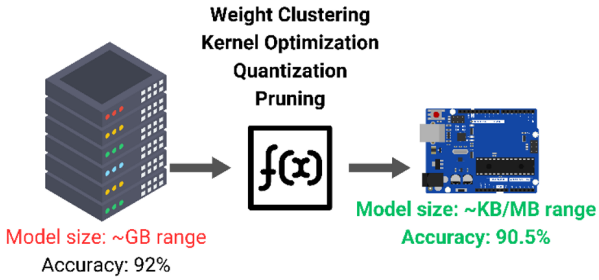


Fig. 3. Model reduction and optimization for edge AI

$$W_i \leftarrow \begin{cases} W_i, & \text{if } \|W_i\|_2 \geq \tau \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

Here W_i denotes the i-th *group* (an output channel in Conv1D/TCN layers or a hidden unit in Dense/RNN layers); groups below the threshold are pruned and the corresponding structures are removed.

Following pruning, post-training quantization (PTQ) was applied to convert all model weights and activations from 32-bit floating-point to 8-bit integers. TensorFlow Lite's full integer quantization pipeline was used, with representative feature vectors provided during calibration. This process generates optimized models suitable for deployment on integer-inference hardware. The optimized models were deployed and evaluated on an NVIDIA Jetson Nano edge device. The Jetson Nano features a quad-core ARM Cortex-A57 CPU and a 128-core Maxwell GPU, making it suitable for AI inference at the edge with moderate power consumption. TensorFlow Lite with the XNNPACK CPU delegate (INT8) and the GPU delegate (FP16) was used for runtime execution.

To assess the practicality and performance of the optimized models on edge hardware, real-time inference testing is performed to ensure the deployed model functions reliably on the Jetson Nano when presented with live or buffered sensor inputs. This step verifies the end-to-end system integration, confirming that the model loads correctly, processes input data without errors, and generates meaningful outputs under real-world or simulated operating conditions. Following this, latency measurement is carried out to quantitatively evaluate the inference speed of the model. To ensure accurate timing, inference latency was measured directly on the Jetson Nano using high-resolution timers within the Python runtime. The measurement captured the time between input feature ingestion and classification output, averaged across multiple inference cycles. All models were evaluated with pre-extracted sliding window features as input, reflecting realistic deployment where raw sensor signals are pre-processed prior to inference. This methodology ensures that deep learning models originally trained on desktop platforms can be effectively compressed and deployed on embedded edge hardware such as the Jetson Nano, enabling intelligent predictive maintenance directly at the source of data collection.

E. Case Severity Analysis for Decision-Making

To augment fault classification with actionable insight, we integrate a two-stage pipeline that includes unsupervised severity estimation of the motor fault. This design enables not only identification of the fault type but also quantification of its relative impact, allowing for more informed maintenance planning. In the first stage, the system performs supervised classification using the trained model to detect the fault type based on input feature vectors, as detailed in III-C. Once a fault is identified, a second, unsupervised analysis is triggered to determine its severity. This severity estimation is based on clustering within a multi-dimensional feature space constructed from the same time-domain and frequency-domain descriptors used in the classification model. These include signal energy (RMS), variability (standard deviation), spectral peaks, crest factor, and harmonic indicators—features that naturally scale with fault intensity. Each fault type (e.g., rotor bar defect, stator fault) is clustered independently to account for differences in how severity manifests across fault categories. Fig. 4 illustrates the end-to-end deployment workflow of the proposed edge AI framework for motor fault diagnosis. Feature vectors computed from current and vibration streams are transmitted wirelessly using Bluetooth Low Energy (BLE) and MQTT protocols to a constrained edge device. The optimized model running on the device first classifies the fault type and subsequently performs severity clustering. Based on the output, real-time alerts and actionable insights can be generated for maintenance or monitoring dashboards.

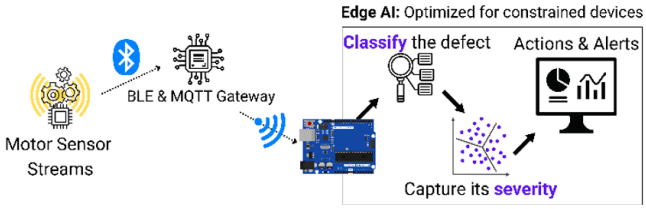


Fig. 4. Edge AI workflow: enabling on-device fault classification and severity analysis for real-time alerts.

We employ K-Means clustering due to its low computational footprint and its suitability for normalized, approximately spherical data distributions [24]. The algorithm partitions the historical feature vectors of a particular fault type into three clusters. The clustering is performed separately for each fault type, using only historical samples previously labeled with that fault. This ensures that differences in severity progression across different fault modes (e.g., rotor vs. stator faults) do not interfere with intra-fault analysis.

Each cluster discovered within this fault-specific feature space is then heuristically labeled as Low, Medium, or High severity, based on the average magnitude of energy-related features within the cluster centroid. This interpretation is data-driven rather than hard-coded, allowing flexible adaptation across deployments. The clustering is performed offline during model development, and the resulting cluster centroids are

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retained for severity inference during deployment. Once a new instance is classified as a fault, its feature vector is assigned to the nearest centroid to estimate its severity. This unsupervised clustering approach eliminates the need for manually defined severity thresholds. It employs a two-tiered decision system, making it suitable for real-time predictive maintenance applications on edge devices. The block diagram of the complete methodology for the fault diagnosis, severity analysis, and predictive maintenance is shown in Fig. 5.

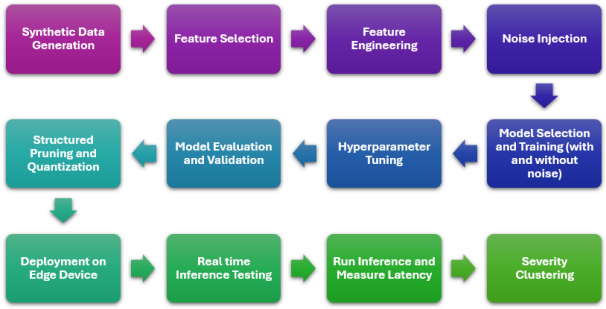


Fig. 5. Block diagram of the proposed methodology

To ensure adaptability in real-world scenarios, the proposed framework is designed with modularity in mind, allowing straightforward retraining or fine-tuning on real datasets. Since the synthetic signals were generated to mimic key physical and statistical properties of real motor faults, the trained models serve as a strong initialization point. Transfer learning techniques can be applied to fine-tune the existing models on small amounts of labeled real-world data, significantly reducing the need for extensive retraining. Additionally, the feature extraction and severity clustering modules are fully data-driven, allowing them to adapt easily as the underlying distribution shifts due to domain-specific variations.

IV. Experimental Results

The experimental evaluation was conducted to analyze the classification accuracy, computational efficiency, and robustness of all five models under varying noise conditions. The models were benchmarked on the high-fidelity synthetic dataset described earlier, across two distinct scenarios: low-noise conditions and severely noisy industrial settings.

A. Model Performance Under Varying Noise Levels

Under low-noise conditions, all models exhibit strong performance, reflecting reliable classification as shown in Fig. 6. The LSTM model achieved the highest macro F1-score of 96%, demonstrating its ability to leverage temporal dependencies for accurate fault diagnosis. Its gated recurrent structure allows it to model sequential patterns and subtle variations in time-series signals, making it highly effective for identifying both abrupt and gradual fault evolution.

Interestingly, Random Forest (RF) achieved a strong macro F1-score of 94%, comparable to deep learning models like RNN (92%), TCN (89%), and GRU (88%). This can be attributed to the clean signal conditions as in the absence of noise, the handcrafted statistical features extracted from the current signals are highly separable across fault classes. Since RF excels at handling structured, tabular features and can learn complex non-linear decision boundaries, it performs competitively when no significant temporal degradation is present. Furthermore, its ensemble nature helps it generalize well without overfitting, especially in low-noise environments.

When subjected to severe noise conditions, including elevated jitter, transient drifts, and voltage fluctuations—the classification performance of all models declined due to increased signal distortion. However, the LSTM model again demonstrated superior robustness, maintaining a macro F1-score of 87%, significantly outperforming other models. In contrast, Random Forest exhibited a sharp decline in performance, dropping to macro F1-score of 70%. Since RF relies solely on static statistical features extracted from each window without modeling any temporal context, its discriminative power degrades significantly when those features become ambiguous or masked by noise. This highlights a key limitation of non-temporal models in real-world deployments, where environmental fluctuations are common. Other sequence models like GRU and TCN also performed moderately well, achieving macro F1-scores of 80% and 81% respectively as represented in Fig. 6

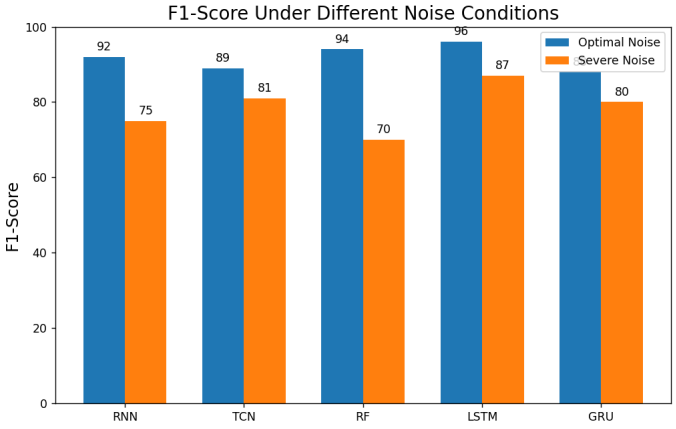


Fig. 6. Performance of models under low-noise and severe-noise conditions

The most pronounced performance degradation under severe noise was observed for mechanical faults, which became more difficult to distinguish due to overlapping features and elevated distortion. To further illustrate this, Fig. 7 presents the confusion matrices of the LSTM model which we chose for its consistently superior performance among all models under both low and severe noise conditions. While the LSTM maintains high true positive rates across most classes in clean conditions, severe noise introduces significant misclassifications, particularly in rotor bar, stator winding and mechanical fault categories, emphasizing the challenge of robust fault detection in noisy environments.

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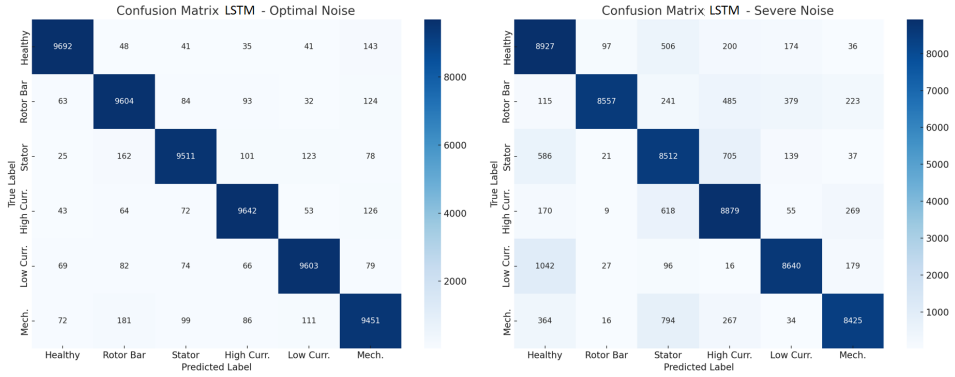


Fig. 7. Confusion Matrix of LSTM with low noise in data (left) and with severe noise in data (right)

B. Model Compression and Embedded Latency Analysis

In this analysis, only the deep learning models (LSTM, GRU, TCN, and RNN) were subjected to compression and optimization for embedded deployment. Random Forest was excluded as it is a tree-based model, and standard compression techniques such as pruning and quantization are tailored for differentiable, tensor-based architectures. Due to its non-differentiable structure and limited compatibility with embedded inference toolchains, RF offers minimal benefits from such optimization and lacks hardware acceleration support on most edge devices.

Table III shows the original and compressed model sizes along with inference latency measured before and after applying quantization and pruning for embedded deployment. Among all models, the LSTM achieved a strong compression ratio (6.66×) and an embedded size of 0.0282 MB, with a latency of 30.8 ms. In contrast, while the TCN achieved the highest compression rate (7.09×), its final size (0.0604 MB) remained the largest among all due to its initially large architecture. Nonetheless, TCN exhibited the lowest embedded latency (25.1 ms), highlighting its efficiency in real-time applications.

TABLE III
ORIGINAL AND COMPRESSED SIZES OF THE MODELS TRAINED ON NOISY DATA WITH INFERENCE LATENCY

Model	Original Size (MB)	Embedded Size (MB)	Original Latency (ms)	Embedded Latency (ms)
LSTM	0.1879	0.0282	36.11	30.8
GRU	0.1490	0.0250	35.54	31.2
TCN	0.4283	0.0604	35.15	25.1
RNN	0.0660	0.0134	35.12	30.1

Table IV further quantifies the impact of compression on macro F1-score and latency. TCN experienced the smallest drop in macro F1-score (only 0.7%) and the largest latency improvement (28.5%), owing to its parallelizable dilated convolutions and stable receptive field design. Despite this, LSTM maintained the highest overall macro F1-score post-compression with only a 0.8% drop, reaffirming its robustness and reliability for noise-resilient embedded inference tasks.

TABLE IV
IMPACT OF MODEL COMPRESSION ON MACRO F1-SCORE AND INFERENCE LATENCY (UNDER SEVERE NOISE)

Model	Compression Rate	Change in macro F1 (%)	Change in Latency (%)
LSTM	6.66×	-0.8	-14.7
GRU	5.96×	-2.1	-12.2
TCN	7.09×	-0.7	-28.5
RNN	4.92×	-1.4	-14.2

C. Case Severity Analysis Results for Decision-Making

After fault classification, we performed unsupervised severity assessment using K-Means clustering on historical samples of each detected fault type. The clustering was conducted independently per fault class in a multi-dimensional feature space combining time-domain (e.g., RMS, kurtosis) and frequency-domain features (e.g., dominant frequency, harmonic energy ratios). Each class was partitioned into three severity levels: low, medium, and high. Although K-Means assumes spherical cluster shapes, silhouette scores exceeding 0.75 for most fault types suggest coherent intra-cluster similarity and sufficient separation. While stator winding faults showed slightly reduced cohesion, the overall segmentation effectively reflected progressive fault behavior. K-Means is selected for its computational efficiency. However, non-spherical clusters may occur in practice. Methods such as DBSCAN or Gaussian Mixture Models could further capture elongated or irregular severity distributions, which is planned to be explored in future work.

To illustrate, Fig. 8 presents the rotor bar fault clustered in 3D space using acceleration RMS, current standard deviation, and current RMS. Despite minor visual overlaps, the achieved silhouette score of 0.77 confirms well-formed clusters in the full feature space. These results support the feasibility of integrating severity estimation in real-time fault monitoring systems.

This two-stage diagnostic pipeline fault classification followed by severity-level clustering, offers interpretable and

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actionable insights. For instance, multiple motors may show the same fault class but differ in their severity cluster due to underlying signal degradation. Embedding this capability into edge-deployable systems can enable condition-based prioritization and early intervention in predictive maintenance workflows. Such severity analysis allows industrial teams to prioritize inspections for high-severity cases, schedule preventive interventions for medium levels, and monitor low-severity instances, thereby optimizing resource allocation and minimizing unplanned downtime.

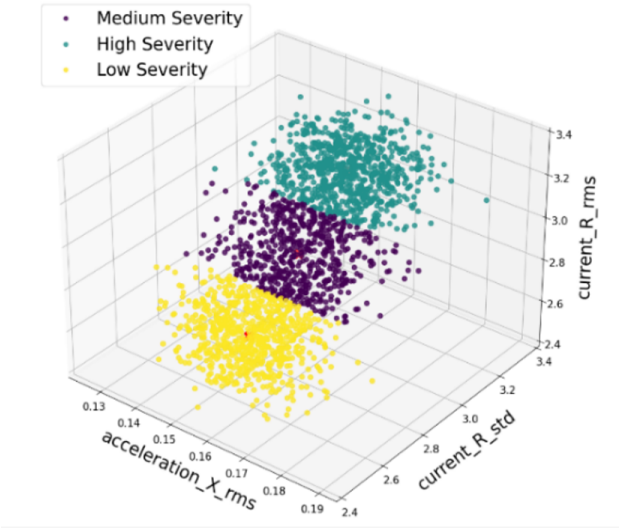


Fig. 8. 3D representation of K-Means clustering for rotor bar defect based on selected features. Average silhouette score is 0.77.

V. Conclusion

This study presents a comprehensive and deployable framework for fault diagnosis and severity assessment in industrial three-phase induction motors using synthetic high-fidelity signal modeling, robust machine learning, and embedded system optimization. An end-to-end framework is introduced from data generation and feature extraction to model training, evaluation under varying noise conditions, and real-time deployment on constrained edge devices.

By simulating six critical electromagnetic and mechanical fault types under both low and severe industrial noise conditions, we demonstrate that sequence-based deep learning models, particularly LSTM, offer superior performance and robustness over classical approaches. To enable deployment in practical industrial environments, we applied structured pruning and post-training quantization techniques, achieving significant reductions in model size and inference latency without substantial loss in accuracy. Our experiments on the Jetson Nano edge platform confirm the feasibility of real-time classification under constrained resources. In addition to classification, a lightweight, unsupervised severity analysis stage using fault-specific clustering in the time–frequency feature space is also described. This augmentation enables more granular, interpretable insights into fault progression, facilitating condition-based maintenance prioritization. The full pipeline—covering classification, severity estimation, and edge deployment—offers a scalable, low-latency solution suitable for modern predictive maintenance systems in smart manufacturing and industrial IoT settings.

Future work will focus on incorporating hybrid signal sources (e.g., thermal, acoustic) and applying domain adaptation strategies and transfer learning techniques to fine-tune the current models on real-world data. Since the synthetic setup closely mimics real motor behavior, the existing models can be adapted with minimal retraining. It is also aimed to validate the framework on actual hardware setups to ensure practical deployment feasibility.

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